



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

Follow For
More Updates





Time Allowed : 2.30 Hours]

MATHEMATICS

[Max. Marks : 100

PART - I

I. Choose the correct Answer.

5x1=5

1. Which of the following is greatest?

a) $\frac{-17}{24}$

b) $\frac{-13}{6}$

c) $\frac{7}{-8}$

d) $\frac{-31}{32}$

2. The number of digits in the square root of 123454321 is _____

a) 4

b) 5

c) 6

d) 7

3. A Cuboid has _____ faces.

a) 4

b) 5

c) 6

d) 7

4. Missing terms of $-3m^2n \times 9(\text{-----}) = \text{-----}m^4n^3$

a) $mn^2, 27$

b) $m^2n, 27$

c) $m^2n^2, -27$

d) $mn^2, -27$

5. One factor of x^3+y^3 is _____

a) $(x-y)$

b) $(x+y)$

c) $(x+y)^3$

d) $(x-y)^3$

II. Fill in the Blanks.

5x1=5

6. The value of x in $x + 5 = 12$ is _____

7. Sum of a number and its half is 30, then the number is _____

8. $(0, -5)$ lies on _____ axis.

9. The range of the data 200, 15, 20, 103, 3, 197 is _____

10. The medians of a triangle cross each other at _____

III. Match the following

5x1=5

11. $\frac{x}{2} = 10$

- $\frac{1}{2} \times d_1 \times d_2$

12. Origin

- $x = 20$

13. Area of Rhombus

- $a^2 + 2ab + b^2$

14. Additive Identity

- $(0,0)$

15. $(a+b)^2$

- 0

IV. Write True or False:

4x1=4

16. 79570 is not a perfect cube.

17. The value of 2^{-2} is 4

18. In any triangle the centroid and the incentre are located inside the triangle.

19. Two numbers are said to be Co - prime numbers if their HCF is 1.

PART - II

V. Answer any 10 of the following.

10x2=20

20. Find : $\frac{-6}{11} + \frac{8}{11} + \frac{-12}{11}$

21. Find the square root of $\frac{144}{225}$

22. Find the value of i) 4^{-3} ii) $\frac{1}{2^{-3}}$

23. A spinner of radius 7.5 cm is divided into 6 equal sectors. Find the area of each of the sectors.

24. Expand : $(3m + 5)^2$

25. Find the value of x if $2x + 5 = 9$

26. What is 25% of 30% of 400?

27. Factorise : $x^2 + 8x + 16$

28. Can a right triangle have sides that measure 5cm, 12 cm and 13 cm?

29. If a company pays Rs. 6 lakh for 15 workers for 20 days, How much would it head to pay 5 workers for 12 days?

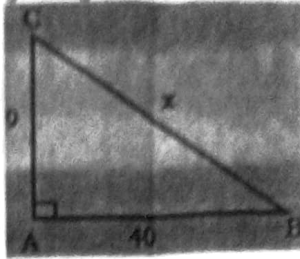
30. Find the compound interest on ₹3200 at 2.5% p.a for 2 years, compounded annually.

CH / 8 / Mat / 1





- 31 Find the unknown side of the following triangle.



- 32 Find the range of the given data.
53, 42, 61, 9, 39, 63, 14, 20, 06, 26, 31, 4, 57.
- 33 Using repeated division method, Find the HCF of 455 and 26.

PART - III

VI. Answer any Nine questions.

- 34 Simplify $\left(\frac{4}{3} - \left(\frac{-3}{2}\right)\right) + \left(\frac{-5}{3} + \frac{30}{12}\right) + \left(\frac{-12}{9} \times \frac{-27}{16}\right)$
- 35 Find the square root of 17956 by long division method.
- 36 The radius of sector is 21 cm and its central angle is 120° , Find its area and Find the length of the arc ($\pi = \frac{22}{7}$)
- 37 Factorise : $49x^2 - 64y^2$
- 38 One number is seven times another of their difference is 18, Find the numbers
- 39 Find the quadrants without plotting the points on a graph
i) (3, -4), ii) (5, 7) iii) (2, 0) iv) (4, -3) v) (0, 10)
- 40 A can do a piece of work in 12 hours, B and C can do it 3 hours whereas A and C can do it in 6 hours. How long will B alone take to do the same work?
- 41 Draw a piechart for the following data relating to the cost of construction of a house.

Particulars	Bricks	Steel	Cement	Timber	Labour	Others
Expenses	10%	15%	25%	10%	20%	20%

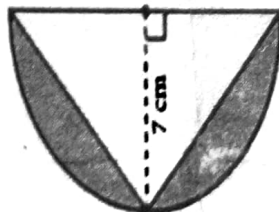
- 42 Find the codes of the following by using Atbash Cipher table.

- (i) GZNR O = QVJML
- (ii) VMTORHS = INLPHUJ
- (iii) NZGSVNZGRXH = AMKJYVW
- (iv) HXRVMXV = WYJQK
- (v) HLXRZOHXRVMXV = YJQKXWJQK

- 43 Draw a histogram for the following data.

Class Interval	0-10	10-20	20-30	30-40	40-50	50-60
No. of students	5	15	23	20	10	7

- 44 Find the area of the shaded portion ($\pi = 3.14$)



- 45 Divide $(5y^3 - 25y^2 + 8y) \div 5y$

PART - IV

VII. Answer all the questions.

2x8=16

- 46 a) Construct a quadrilateral DEAR with DE = 6cm, EA = 5cm, AR = 5.5 cm, RD = 5.2 cm, DA = 10 cm. Find its area. (OR)
- b) Construct a rectangle BEAN with BE = 5cm and BN = 3cm. Also find its area.
- 47 a) Draw the graph of $y = 5x$. (OR)
- b) Plot the following points in the graph sheet.
A (5, 2) B (-7, -3) C (-2, 4) D (-1, -1) E (0, -5) F (2, 0) G (7, -4) H (-4, 0)